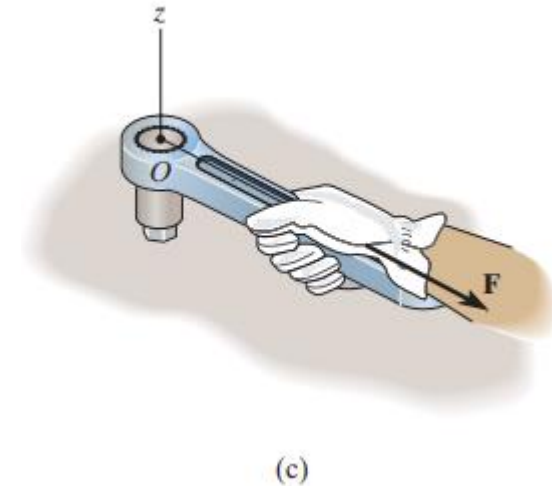
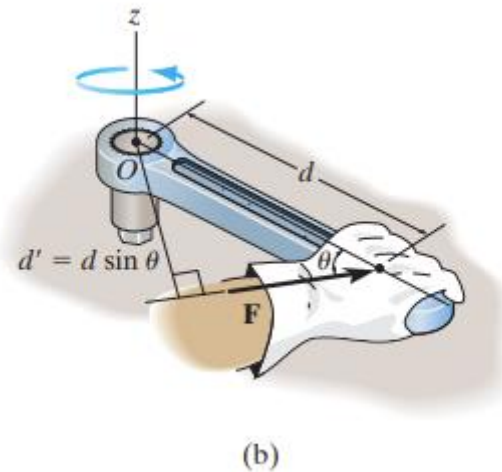
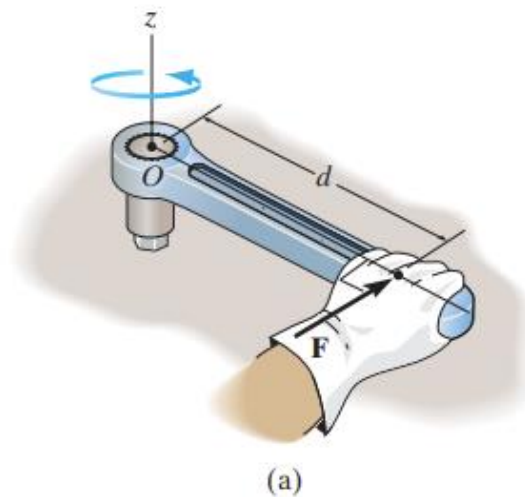


Gebze Technical University
Dep. of Mechanical Engineering

Statics (ME 231)
Lecture 5: Force System
Resultants (Part 1)
By Saeed Lotfan

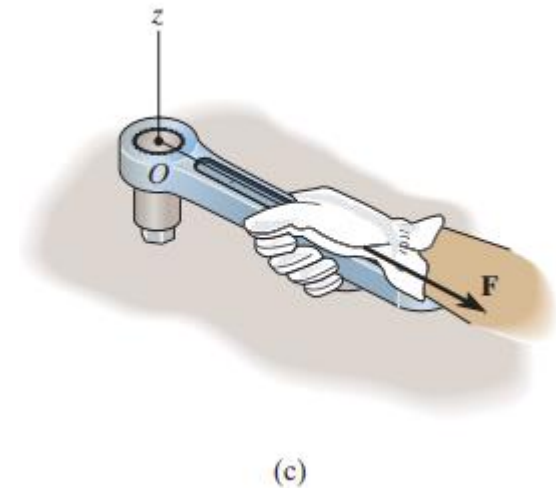
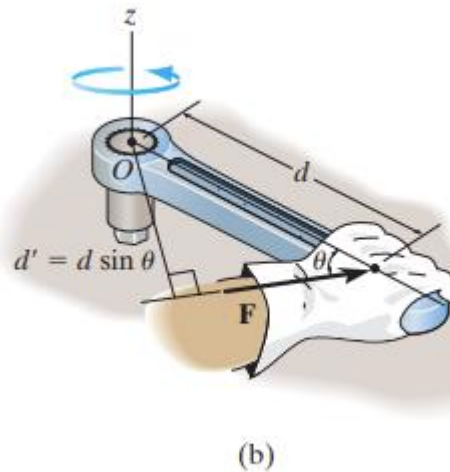
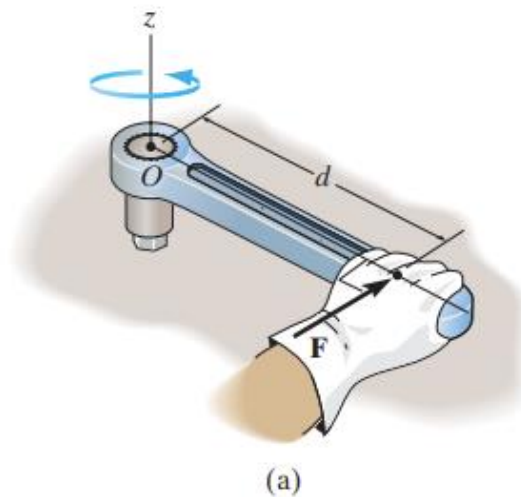
Moment of A Force

When a force is applied to a body it will produce a tendency for the body to rotate about a point that is not on the line of action of the force. This tendency to rotate is sometimes called **torque**, but most often it is called the moment of a force or simply the **moment**.



Moment of A Force

The magnitude of the moment is directly proportional to the magnitude of $\mathbf{F}$ and the perpendicular distance or moment arm d . The larger the force or the longer the moment arm, the greater the moment or turning effect.



Moment of A Force – Scalar Formulation

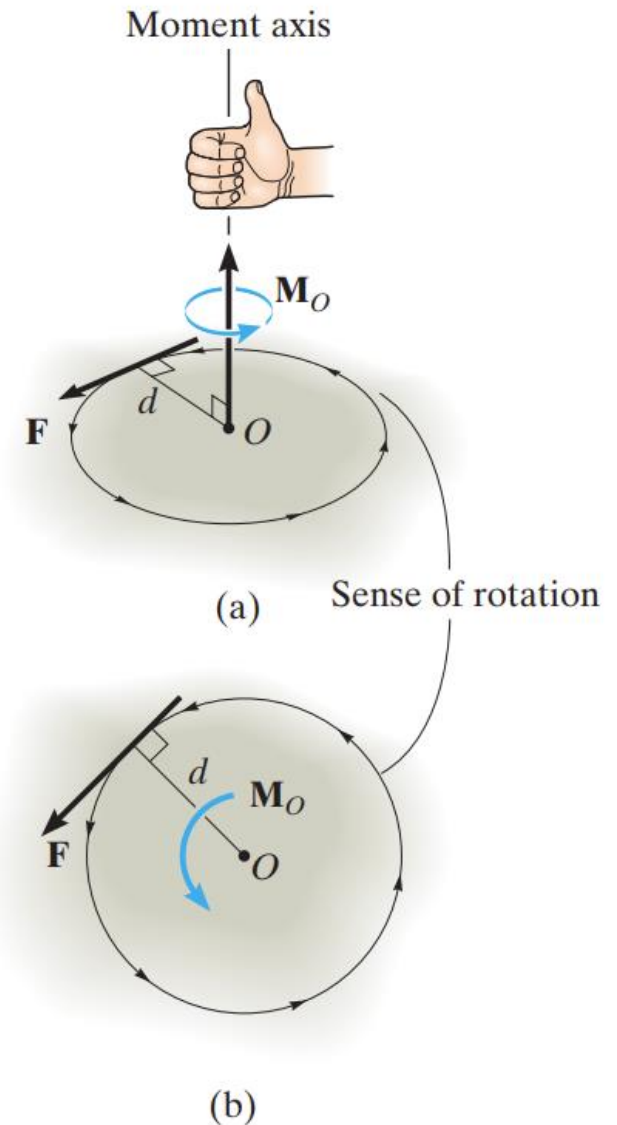
Magnitude

$$M_O = Fd$$

where d is the **moment arm** or perpendicular distance from the axis at point O to the line of action of the force. Units of moment magnitude consist of force times distance, e.g., N.m.

Direction

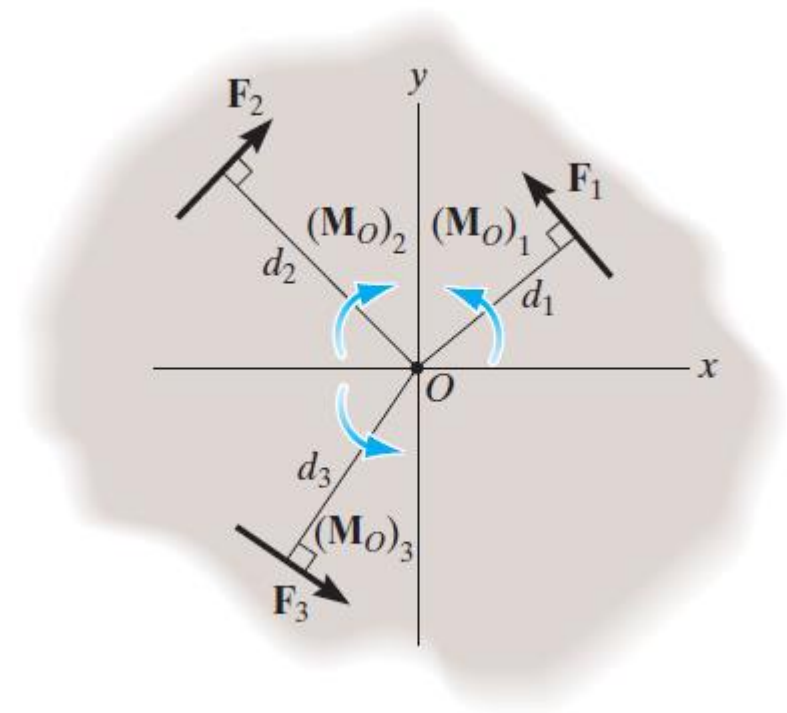
The direction of $\mathbf{M}_O$ is defined by its moment axis, which is perpendicular to the plane that contains the force $\mathbf{F}$ and its moment arm d . The right-hand rule is used to establish the sense of direction of $\mathbf{M}_O$.



Moment of A Force – Scalar Formulation

Resultant Moment

For two-dimensional problems, where all the forces lie within the x – y plane, the resultant moment $(\mathbf{M}_R)_O$ about point O (the z axis) can be determined by finding the algebraic sum of the moments caused by all the forces in the system. As a convention, we will generally consider positive moments as counterclockwise since they are directed along the positive z axis (out of the page). Clockwise moments will be negative.



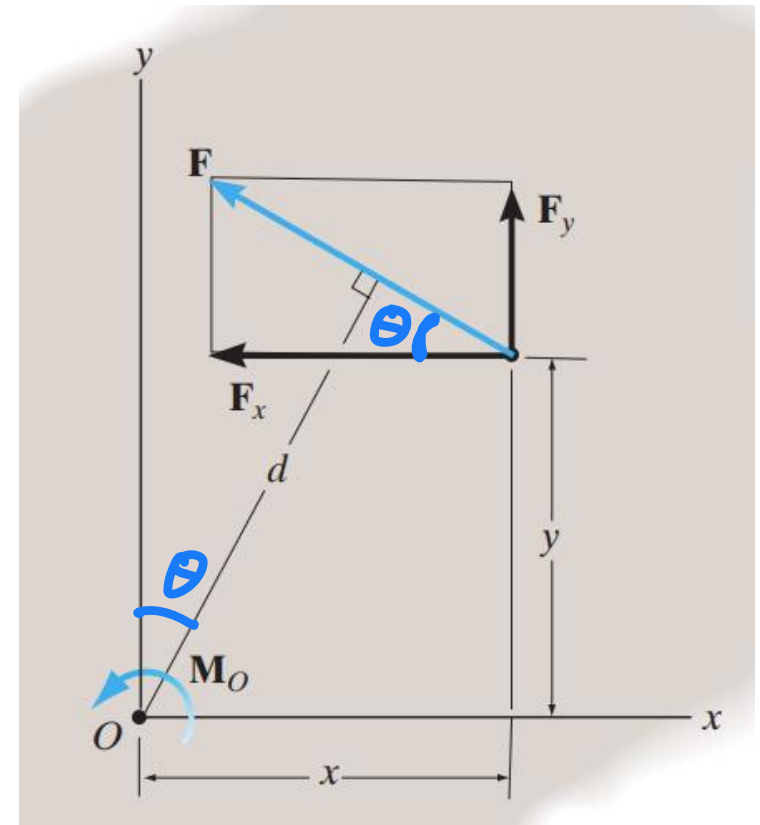
$$\downarrow + (M_R)_O = \Sigma Fd; \quad (M_R)_O = F_1d_1 - F_2d_2 + F_3d_3$$

Principle of Moments (Varignon's theorem)

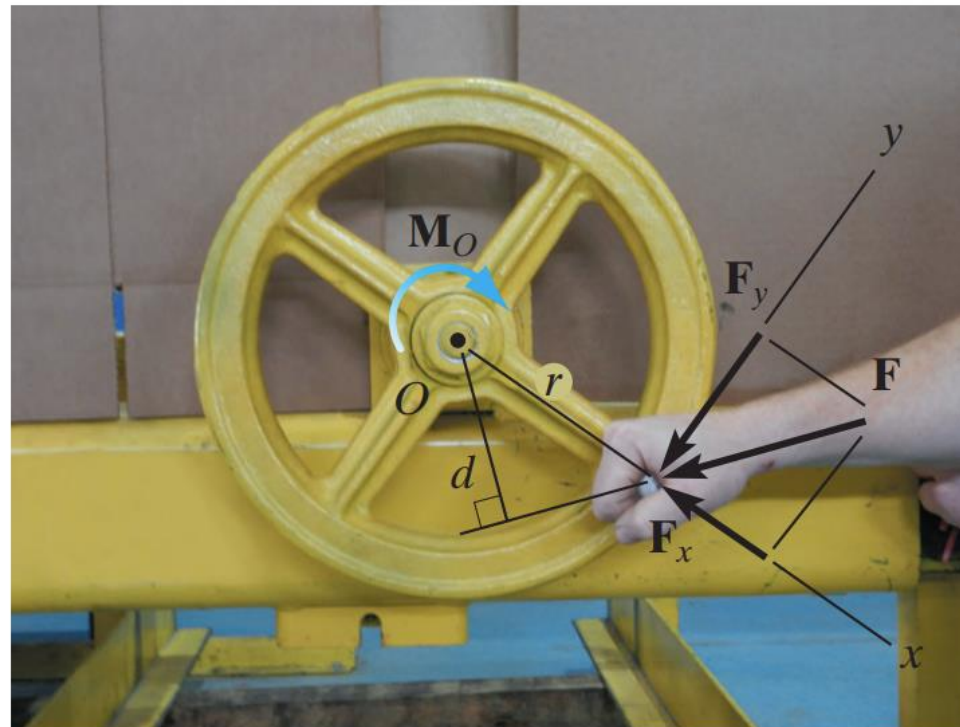
It states that the moment of a force about a point is equal to the sum of the moments of the components of the force about the point.

$$\begin{aligned}
 M_O^2 &= F^2 d^2 \\
 &= F^2 (d^2 \sin^2 \theta + d^2 \cos^2 \theta) \\
 &= F^2 (x^2 + y^2) \\
 M_O &= F_x y + F_y x
 \end{aligned}$$

$$M_O = F_x y + F_y x$$

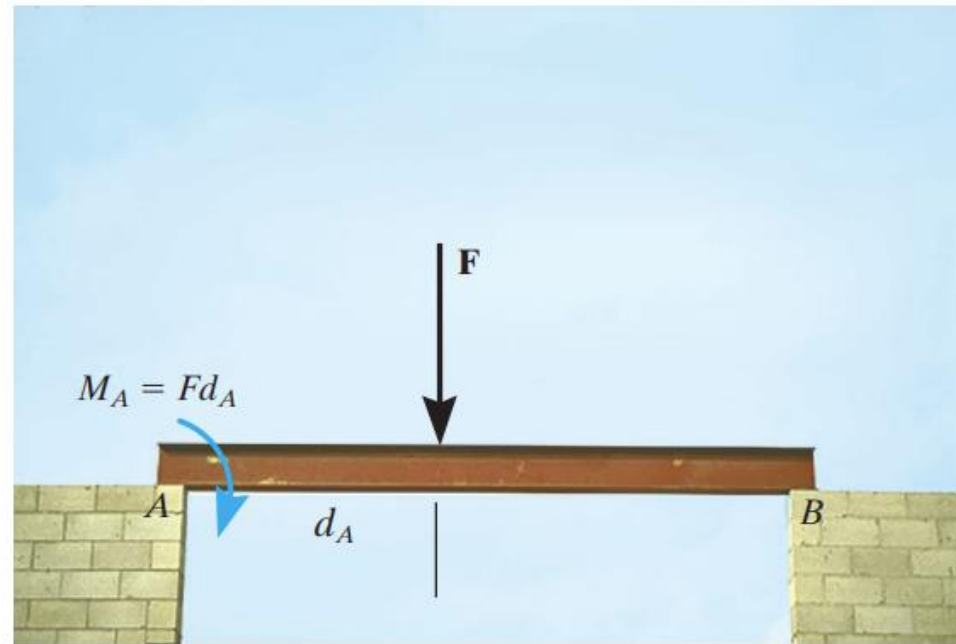


Principle of Moments (Varignon's theorem)



The moment of the force about point O is $M_O = Fd$. But it is easier to find this moment using $M_O = F_x(0) + F_y r = F_y r$.

Principle of Moments (Varignon's theorem)

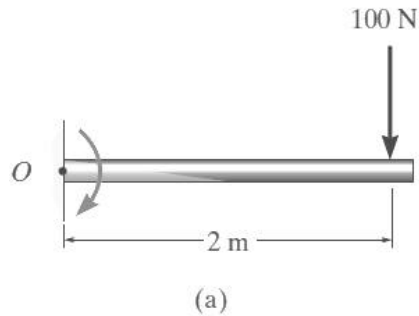


The force $\mathbf{F}$ tends to rotate the beam clockwise about its support at A with a moment $M_A = Fd_A$. The actual rotation would occur only if the support at B were removed.

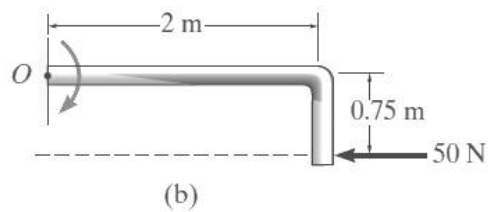
Concluding Notes

- The moment of a force creates the tendency of a body to turn about an axis passing through a specific point O .
- Using the right-hand rule, the sense of rotation is indicated by the curl of the fingers, and the thumb produces the sense of direction of the moment.
- The magnitude of the moment is determined from $M_O = Fd$, where d is called the moment arm, which represents the perpendicular or shortest distance from point O to the line of action of the force.
- In two dimensions it is often easier to use the principle of moments and find the moment of the force components about point O , rather than using $M_O = Fd$.

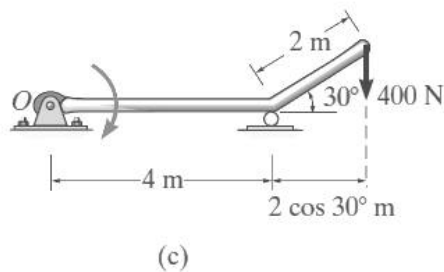
Example: For each case illustrated below, determine the moment of the force about point O.



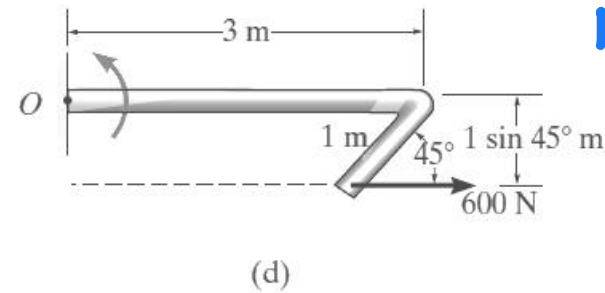
$$\begin{aligned}
 M_O &= Fd \\
 &= 100 \cdot (2) \\
 &= 200 \text{ N}\cdot\text{m} \downarrow
 \end{aligned}$$



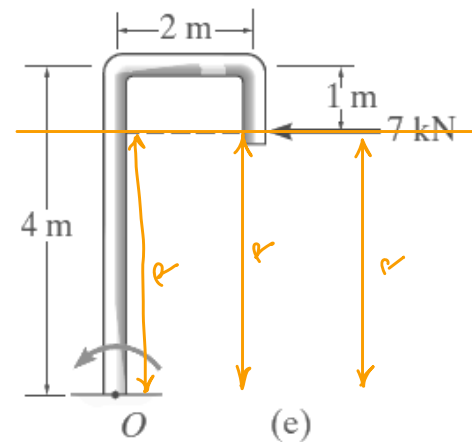
$$M_O = 50 \cdot (0.75) \downarrow$$



$$M_O = 400(4 + 2\cos 30)$$

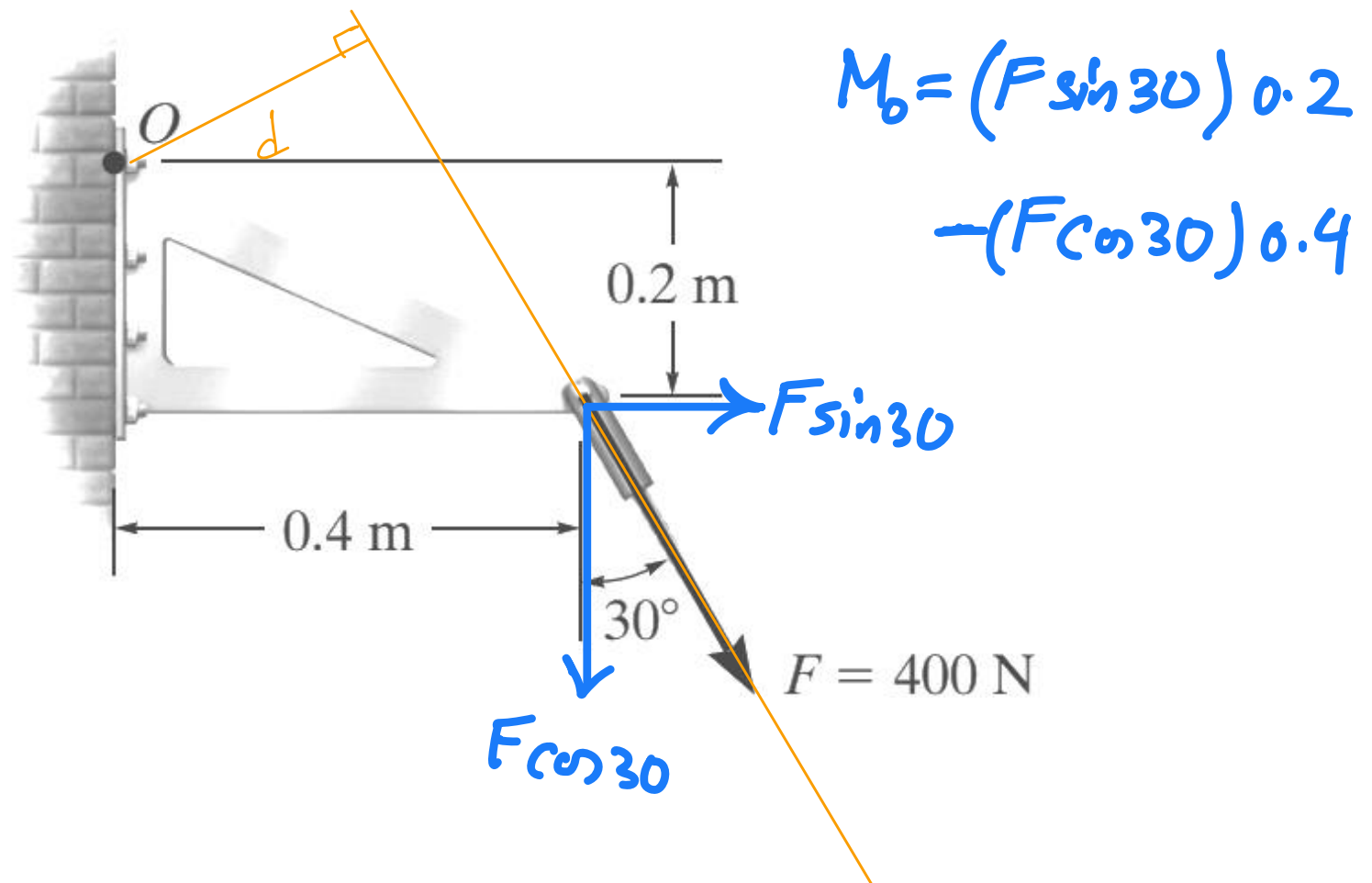


$$M_O = 600(1 \sin 45)$$



$$M_O = 7000(3) \text{ N}\cdot\text{m}$$

Example: Force $\mathbf{F}$ acts at the end of the angle bracket shown below. Determine the moment of the force about point O .



Cross Product

The cross product of two vectors **A** and **B** yields the vector **C**, which is written

$$\mathbf{C} = \mathbf{A} \times \mathbf{B}$$

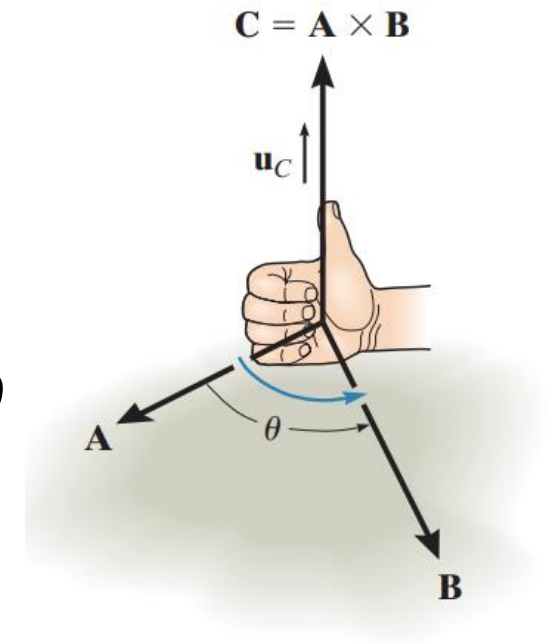
and is read “**C** equals **A** cross **B**.”

Magnitude

The magnitude of **C** is defined as the product of the magnitudes of **A** and **B** and the sine of the angle θ between their tails, $C = AB \sin \theta$.

Direction

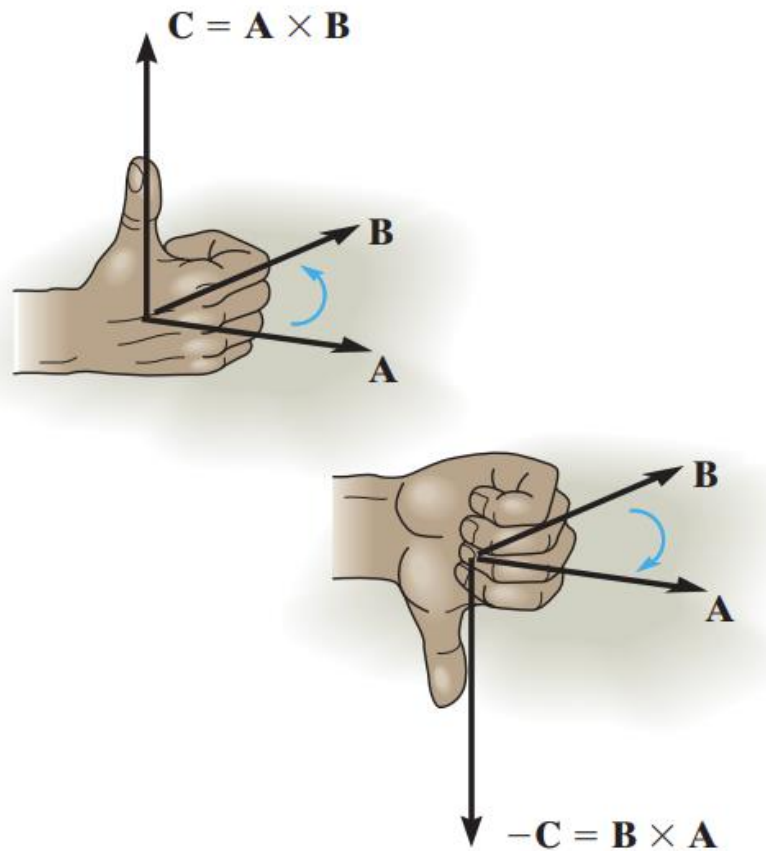
Vector **C** has a direction that is perpendicular to the plane containing **A** and **B** such that the directional sense of **C** is specified by the right-hand rule.



Cross Product

Laws of Operation

Cross product is not commutative!



Cross Product

Laws of Operation

Cross product is both associative and distributive!

$$a(\mathbf{A} \times \mathbf{B}) = (a\mathbf{A}) \times \mathbf{B} = \mathbf{A} \times (a\mathbf{B}) = (\mathbf{A} \times \mathbf{B})a$$

$$\mathbf{A} \times (\mathbf{B} + \mathbf{D}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{D})$$

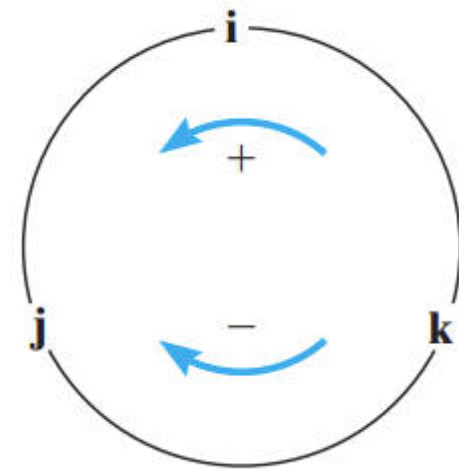
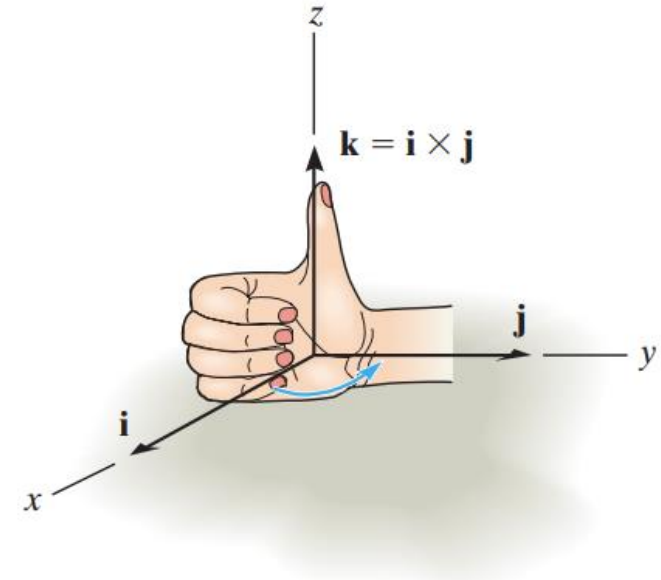
Cross Product

Cartesian Vector Formulation

$$\mathbf{i} \times \mathbf{j} = \mathbf{k} \quad \mathbf{i} \times \mathbf{k} = -\mathbf{j} \quad \mathbf{i} \times \mathbf{i} = \mathbf{0}$$

$$\mathbf{j} \times \mathbf{k} = \mathbf{i} \quad \mathbf{j} \times \mathbf{i} = -\mathbf{k} \quad \mathbf{j} \times \mathbf{j} = \mathbf{0}$$

$$\mathbf{k} \times \mathbf{i} = \mathbf{j} \quad \mathbf{k} \times \mathbf{j} = -\mathbf{i} \quad \mathbf{k} \times \mathbf{k} = \mathbf{0}$$



Cross Product

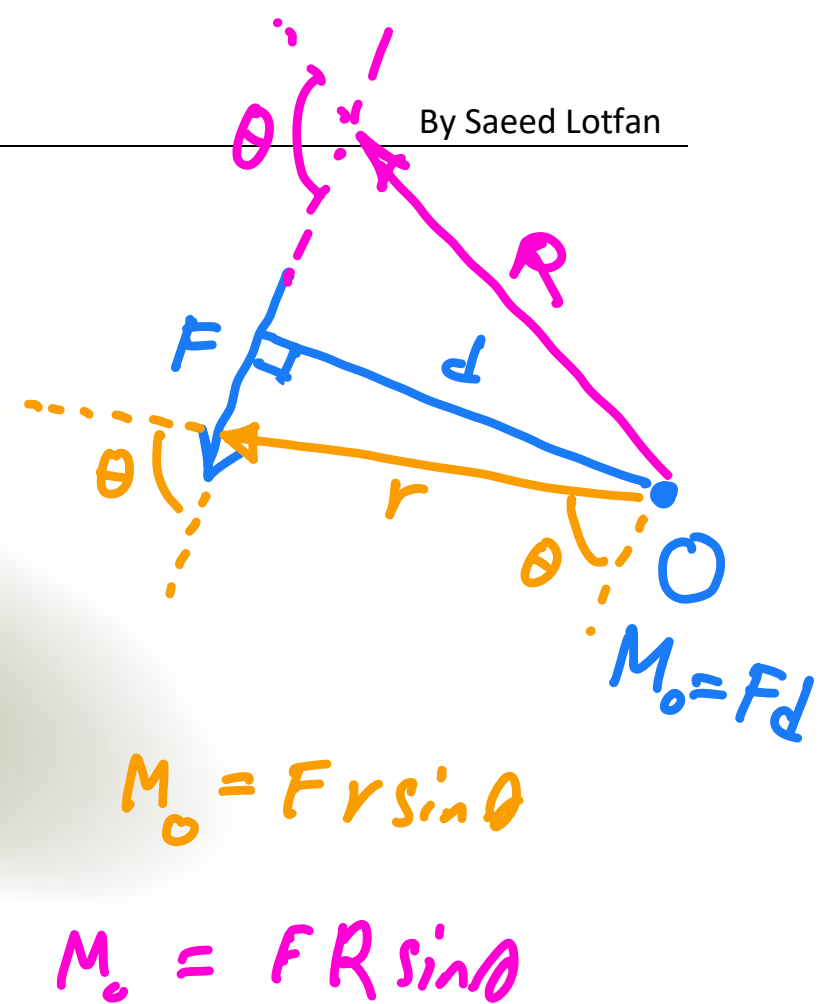
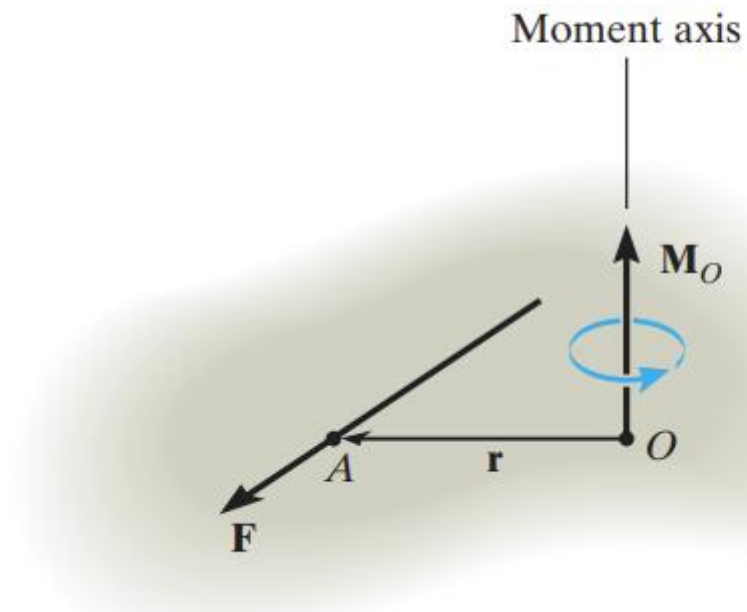
Determinant Formulation

$$\begin{aligned}
 \mathbf{A} \times \mathbf{B} &= (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \times (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) \\
 &= A_x B_x (\mathbf{i} \times \mathbf{i}) + A_x B_y (\mathbf{i} \times \mathbf{j}) + A_x B_z (\mathbf{i} \times \mathbf{k}) \\
 &\quad + A_y B_x (\mathbf{j} \times \mathbf{i}) + A_y B_y (\mathbf{j} \times \mathbf{j}) + A_y B_z (\mathbf{j} \times \mathbf{k}) \\
 &\quad + A_z B_x (\mathbf{k} \times \mathbf{i}) + A_z B_y (\mathbf{k} \times \mathbf{j}) + A_z B_z (\mathbf{k} \times \mathbf{k})
 \end{aligned}$$

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y) \mathbf{i} - (A_x B_z - A_z B_x) \mathbf{j} + (A_x B_y - A_y B_x) \mathbf{k}$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

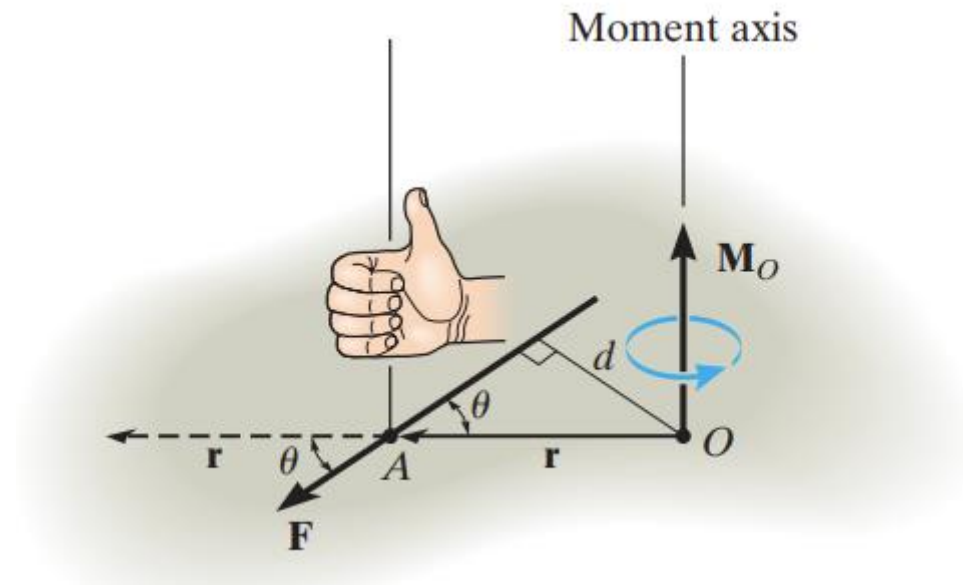
Moment of A Force - Vector Formulation



$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

Moment of A Force - Vector Formulation

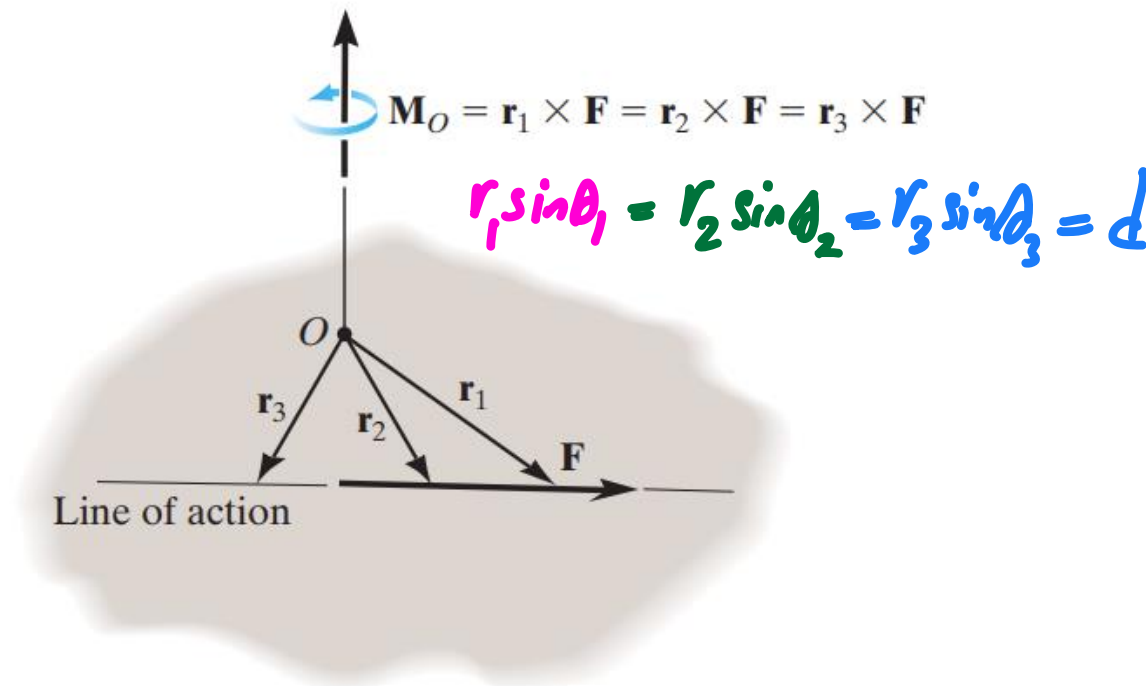
Magnitude and Direction



$$M_O = rF \sin \theta = F(r \sin \theta) = Fd$$

Moment of A Force - Vector Formulation

Principle of Transmissibility of a Force

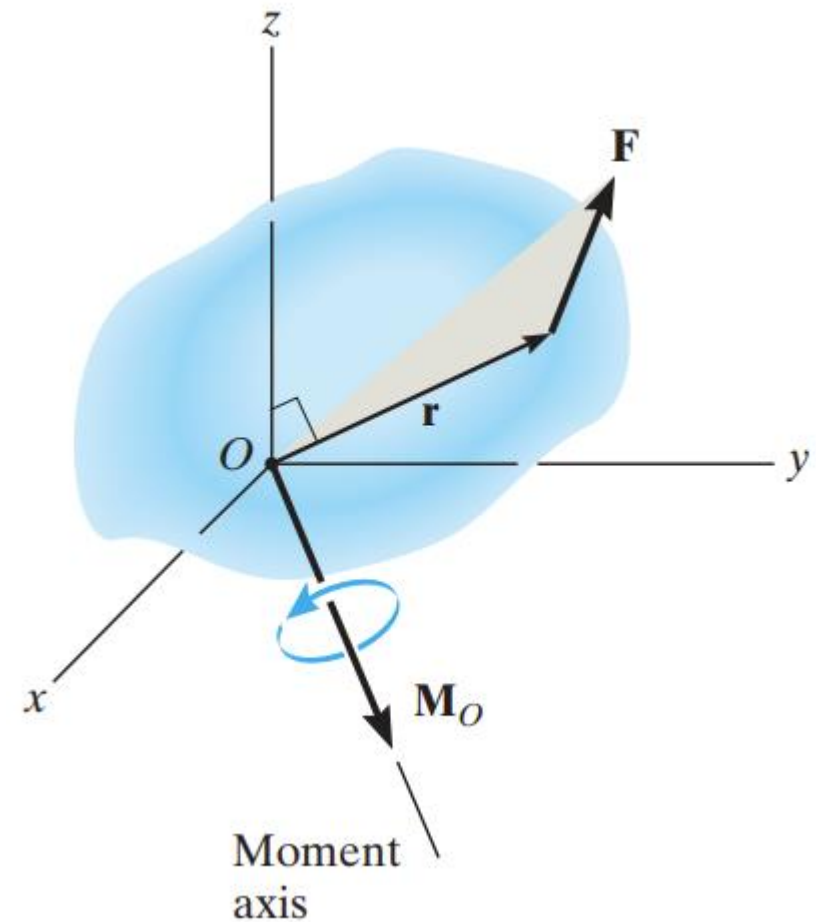


Since $\mathbf{F}$ can be applied at any point along its line of action and still creates this same moment about point O , then $\mathbf{F}$ can be considered a **sliding vector**. This property is called the principle of transmissibility of a force.

Moment of A Force - Vector Formulation

Cartesian Vector Formulation

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

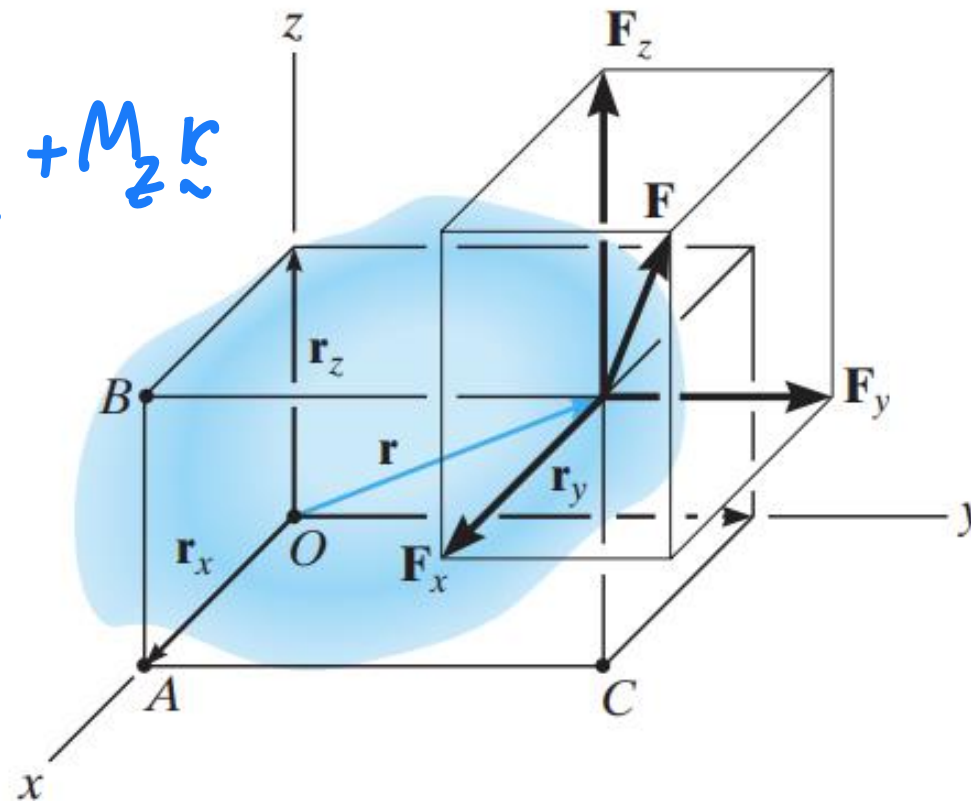


Moment of A Force - Vector Formulation

Physical Meaning of Cartesian Vector Formulation

$$\vec{M}_O = \vec{r} \times \vec{F}$$

$$M_O = M_x \hat{i} + M_y \hat{j} + M_z \hat{k}$$



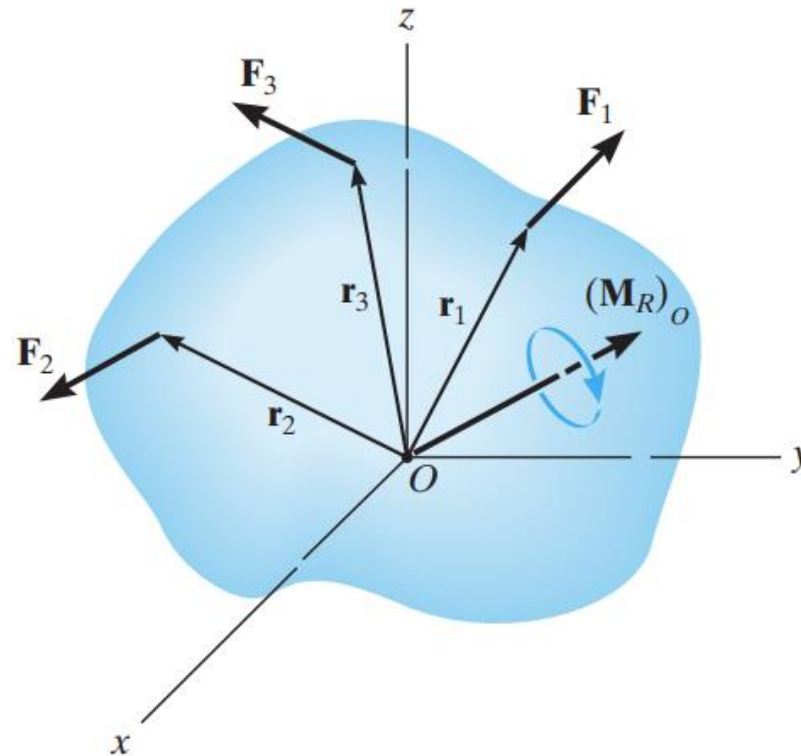
F_x moment

$F_x r_z$ $\rightarrow$ y direction

$F_x r_y$ $\rightarrow$ -z direction

Moment of A Force - Vector Formulation

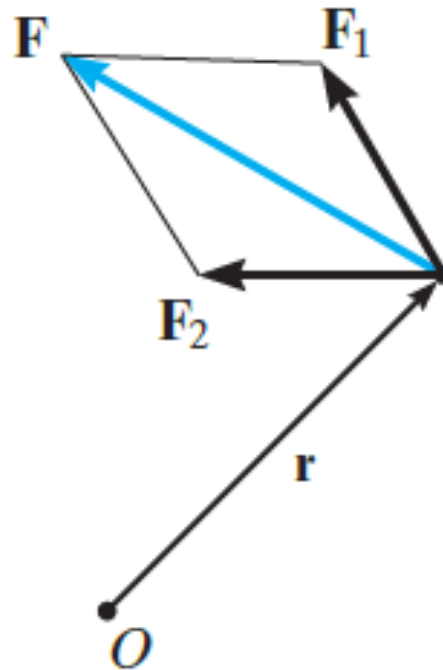
Resultant Moment of a System of Forces



$$(\mathbf{M}_R)_O = \Sigma(\mathbf{r} \times \mathbf{F})$$

Moment of A Force - Vector Formulation

Principle of Moments (Varignon's theorem and distributive law)



$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2) = \mathbf{r} \times \mathbf{F}_1 + \mathbf{r} \times \mathbf{F}_2$$

Example: Determine the moment produced by the force $\mathbf{F}$ shown below about point O . Express the result as a Cartesian vector.

$$16 + 144 + 144 = 304$$

$$\vec{r}_{B/A} = \vec{r}_B - \vec{r}_A = (4\vec{i} + 12\vec{j}) - 12\vec{k}$$

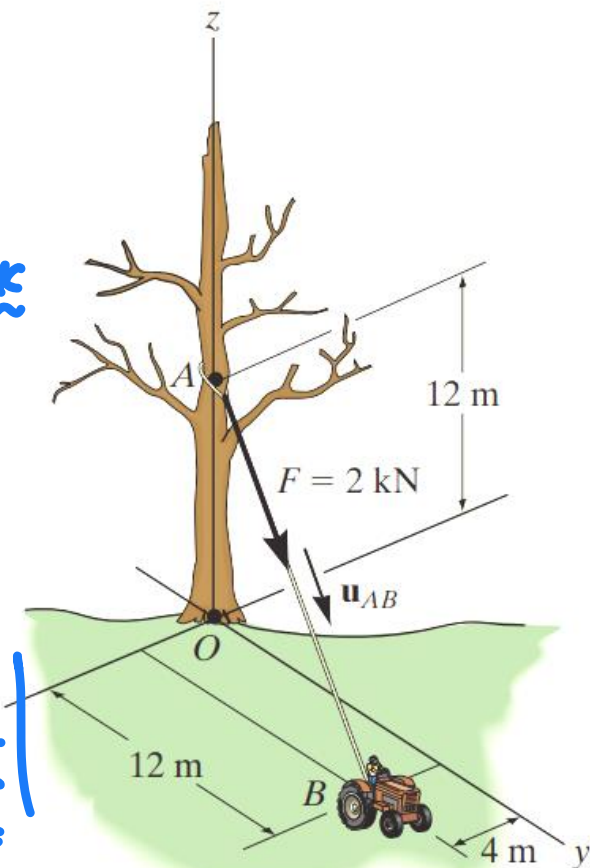
$$\vec{u}_{B/A} = \frac{4}{\sqrt{304}}\vec{i} + \frac{12}{\sqrt{304}}\vec{j} - \frac{12}{\sqrt{304}}\vec{k}$$

$$\vec{F} = 2000 \vec{u}_{B/A} \text{ N}$$

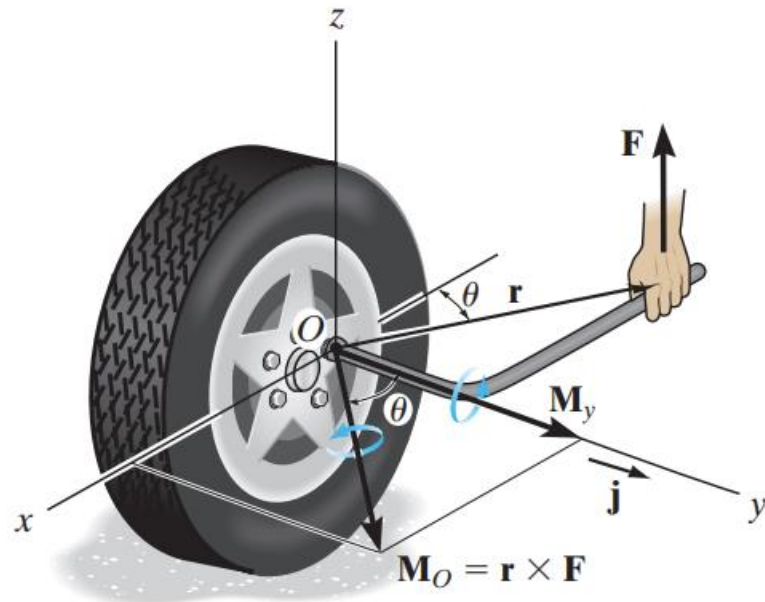
$$\vec{M}_O = \vec{r} \times \vec{F}$$

$$\vec{r} = 12\vec{k}$$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 12 \\ F_x & F_y & F_z \end{vmatrix}$$



Moment of A Force About a Specified Axis



Scalar Analysis

$$M_a = Fd_a$$

Realize that a force will not create a moment about an axis (or turn about an axis) if it is *parallel to the axis*, or if its *line of action passes through the axis*.

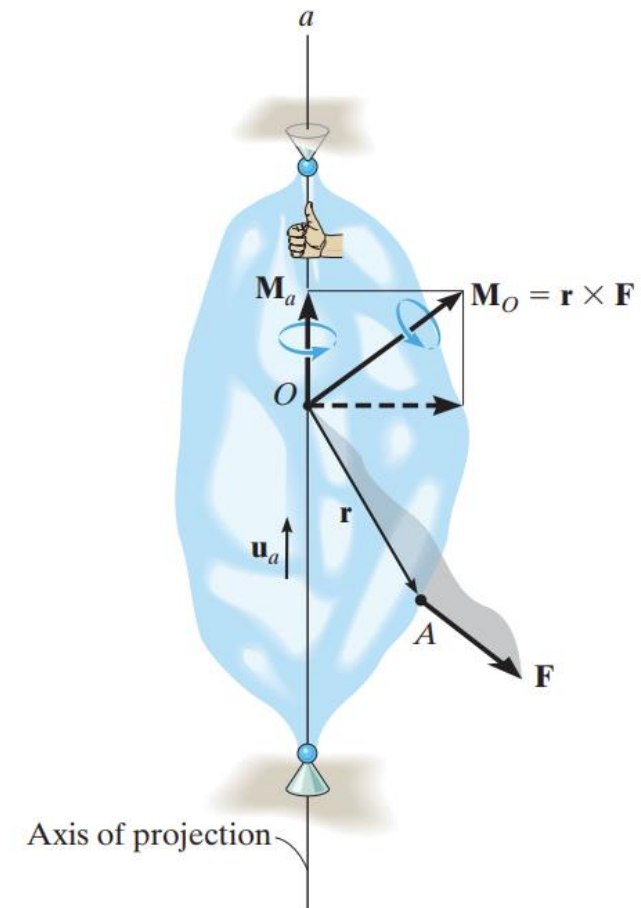
Moment of A Force About a Specified Axis

Vector Analysis

$$M_a = [u_{a_x}\mathbf{i} + u_{a_y}\mathbf{j} + u_{a_z}\mathbf{k}] \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

$$= u_{a_x}(r_y F_z - r_z F_y) - u_{a_y}(r_x F_z - r_z F_x) + u_{a_z}(r_x F_y - r_y F_x)$$

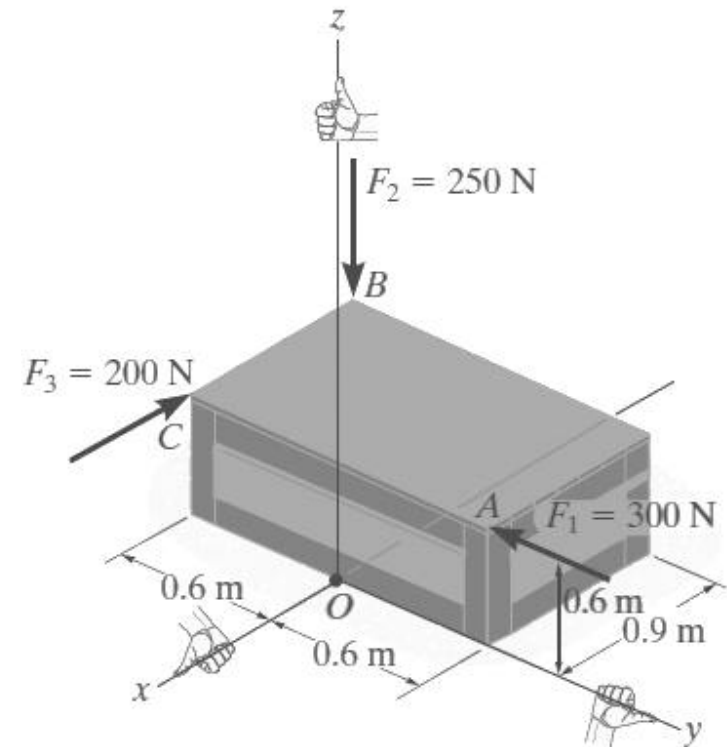
$$\mathbf{M}_a = M_a \mathbf{u}_a$$



Concluding Notes

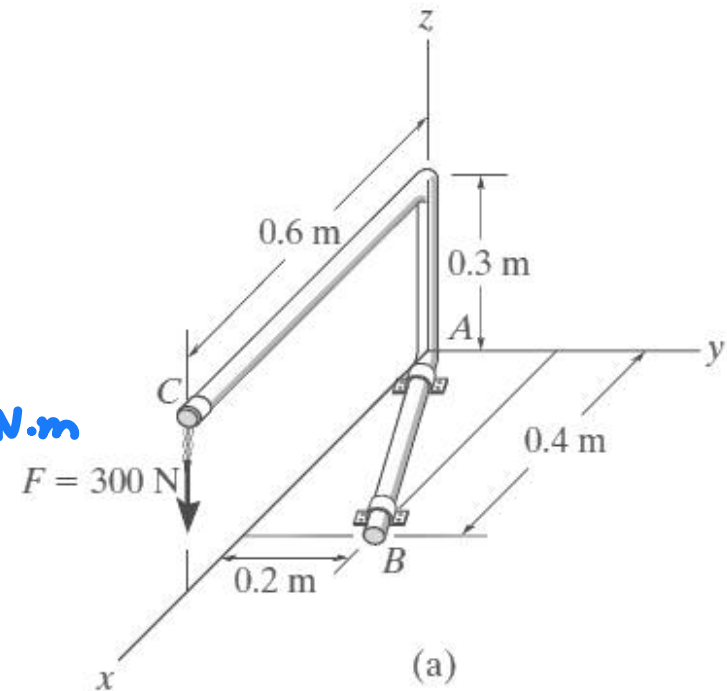
- The moment of a force about a specified axis can be determined provided the perpendicular distance d_a from the force line of action to the axis can be determined. Then $M_a = Fd_a$.
- No moment is produced if the force is parallel to the axis or its line of action passes through it.
- If vector analysis is used, then $M_a = \mathbf{u}_a \cdot (\mathbf{r} \times \mathbf{F})$, where $\mathbf{u}_a$ defines the direction of the axis and $\mathbf{r}$ is extended from *any point* on the axis to *any point* on the line of action of the force.
- If M_a is calculated as a negative scalar, then the sense of direction of $\mathbf{M}_a$ is opposite to $\mathbf{u}_a$.
- The moment $\mathbf{M}_a$ expressed as a Cartesian vector is determined from $\mathbf{M}_a = M_a \mathbf{u}_a$.

Example: Determine the resultant moment of the three forces shown below about the x axis, the y axis, and the z axis.



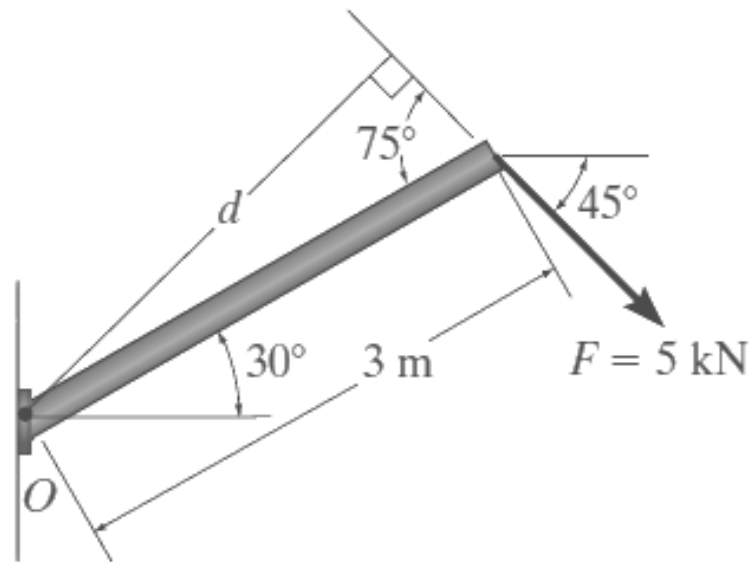
Example: Determine the moment $\mathbf{M}_{AB}$ produced by the force $\mathbf{F}$ shown below, which tends to rotate the rod about the AB axis.

$$\begin{aligned} M_A &= \vec{r}_1 \times \vec{F} \\ \vec{F} &= -300 \hat{k} \\ \vec{r}_1 &= 0.6 \hat{i} \\ \vec{M}_A &= 180 \hat{j} \text{ N}\cdot\text{m} \end{aligned}$$



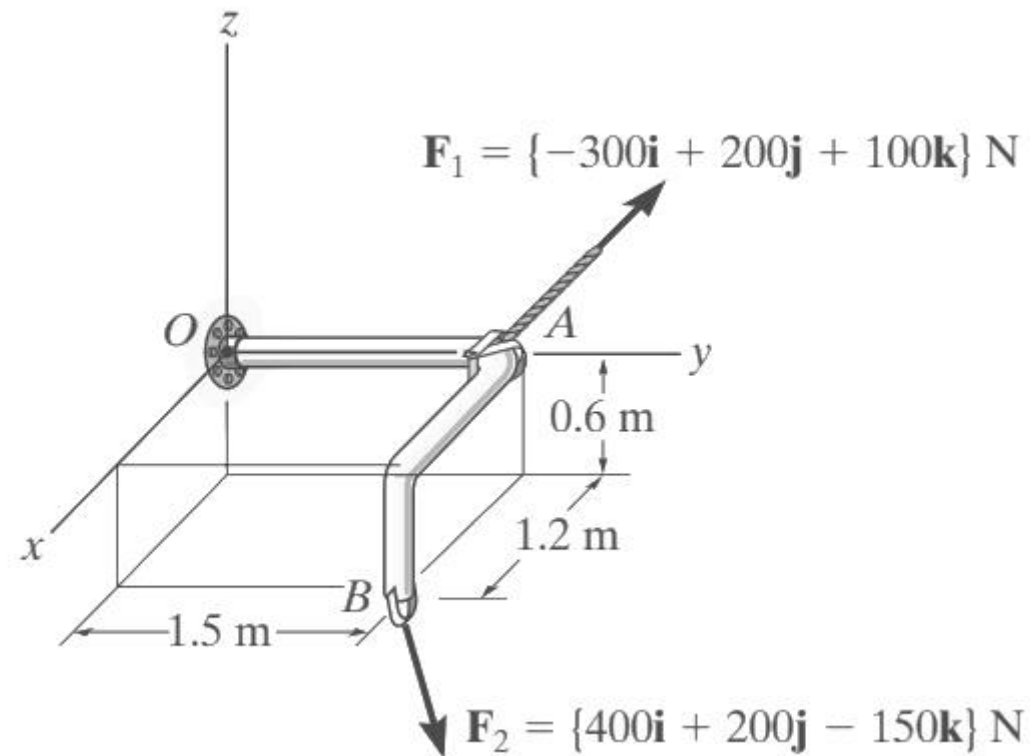
Practice at Home

1. Determine the moment of the force shown below about point O .



Practice at Home

2. Two forces act on the rod shown below. Determine the resultant moment they create about the flange at O. Express the result as a Cartesian vector.



Practice at Home

3. Determine the magnitude of the moment of force **F** about segment OA of the pipe assembly in the following figure.

$$\vec{r} = 0.1\vec{i} + 0.4\vec{j} + 0.3\vec{k}$$

$$\vec{r}_{D/C} = \vec{r}_D - \vec{r}_C$$

$$= (0.5\vec{i} + 0\vec{j} + 0.5\vec{k})$$

$$- (0.1\vec{i} + 0.4\vec{j} + 0.3\vec{k})$$

$$\vec{r}_{D/C} = (0.4\vec{i} - 0.4\vec{j} + 0.2\vec{k})$$

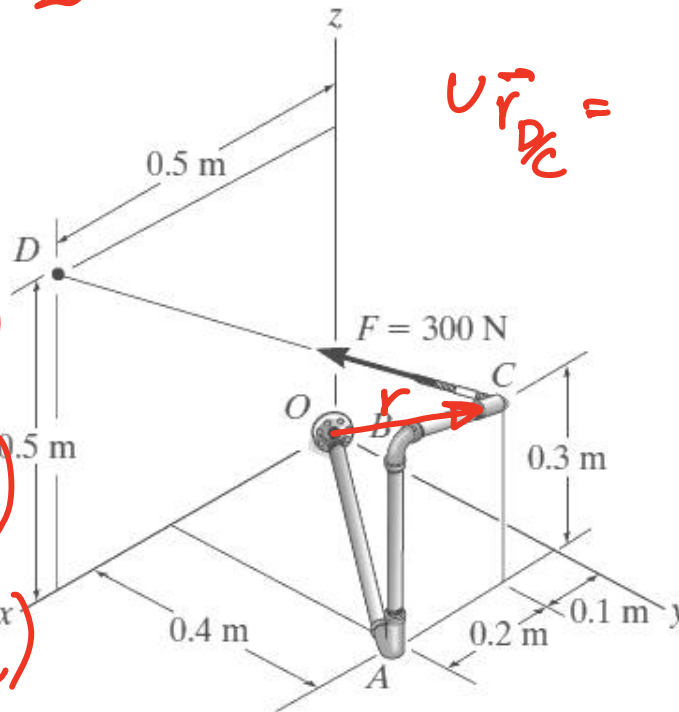
$$\vec{F} = 300(u_x\vec{i} + u_y\vec{j} + u_z\vec{k})$$

$$u_{\vec{r}_{D/C}} = \frac{\vec{r}_{D/C}}{|\vec{r}_{D/C}|} = u_x\vec{i} + u_y\vec{j} + u_z\vec{k}$$

$$\vec{r} \times \vec{F} = \vec{M}_A$$

$$\vec{M}_{OA} = \vec{u}_{OA}(\vec{M}_A \cdot \vec{u}_{OA})$$

$$\vec{u}_{OA} = \frac{\vec{r}_{A/O}}{|\vec{r}_{A/O}|}$$



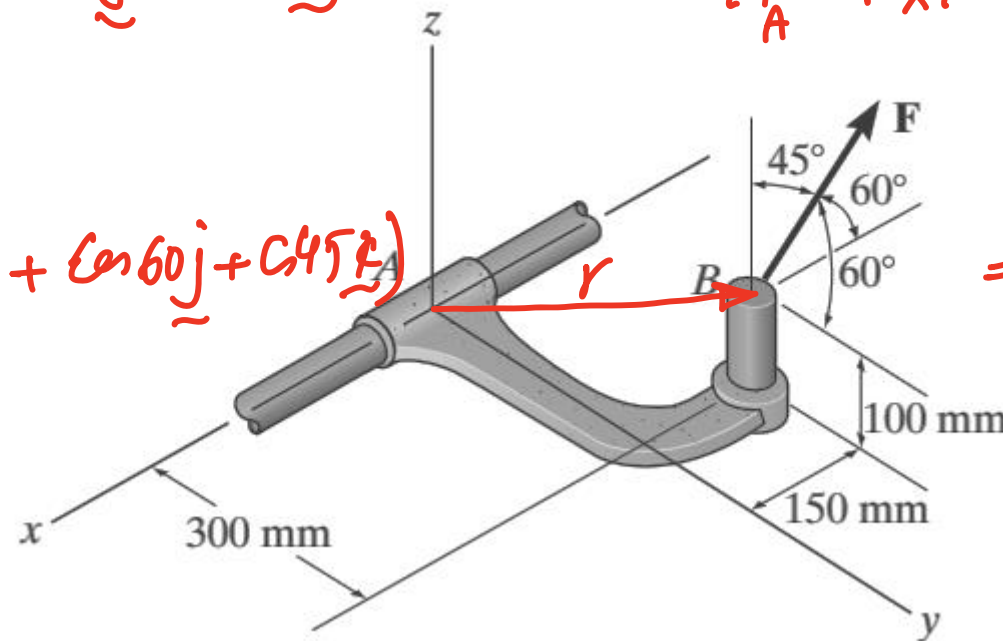
Practice at Home

4. The friction at sleeve A can provide a maximum resisting moment of 125 N.m about the x axis. Determine the largest magnitude of force $\mathbf{F}$ that can be applied to the bracket so that the bracket will not turn.

$$\vec{r} = -150\hat{i} + 300\hat{j} + 100\hat{k}$$

$$\vec{u}_r = \frac{\vec{r}}{|\vec{r}|}$$

$$\vec{F} = F(-\cos 60^\circ \hat{i} + \cos 60^\circ \hat{j} + \cos 45^\circ \hat{k})$$



$$M_A = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 150 & 300 & 100 \\ F\cos 60^\circ & F\cos 60^\circ & F\cos 45^\circ \end{vmatrix} = N_{\text{mm}}$$

$$= M_x \hat{i} + M_y \hat{j} + M_z \hat{k}$$

N.m

$$M_x = 125 \text{ N.m}$$